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Paper Practice 1.5 • WORKED KEY

Absolute Value Expressions

WORKED KEY (odd-numbered items) – Show all of your work in the space provided. Simplify INSIDE the absolute value bars first. Circle your final answers.

1. Let $a = 5$, $b = -9$, and $c = 3$. Evaluate each expression. (Absolute Value)

a. $|b|$

Answer: $|-9| = 9$

b. $-|a|$

Answer: $-|5| = -5$ – the negative is OUTSIDE the bars

c. $|-c|$

Answer: $|-3| = 3$

d. $-|-b|$

Answer: $-|-(-9)| = -|9| = -9$

2. Evaluate each expression. (Evaluate)

a. $|3x + 7|$, if $x = -4$

b. $|-5x - 3y|$, if $x = 2$ and $y = -4$

3. Evaluate $|-3x + 4y|$, if $x = 6$ and $y = -2$. (Evaluate)

Answer: $-18 + (-8) = -26$; $|-26| = 26$

4. Evaluate $18 - |2 + 3x|$, if $x = 4$. (Evaluate)

5. Evaluate $3|4x - 14| + 5$, if $x = 2$. (Evaluate)

Answer: $8 - 14 = -6$; $|-6| = 6$; $3(6) + 5 = 23$

6. Evaluate $2.5 - 1.5|4y + 1.2|$, if $y = -2$. (Evaluate)

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7. The accuracy of a kitchen scale is the positive difference between the reading r and the true weight w. Write TWO expressions that both represent the accuracy. (Write)

Answer: $|r - w|$ and $|w - r|$ – order inside absolute value bars does not matter

8. A scale reads 4.92 pounds for a bag labeled 5 pounds. Use your expression from #7 to find the accuracy. (Apply)

9. Explain in one sentence why $|x|$ can never be negative, no matter what x is. (Explain)

Answer: Absolute value measures DISTANCE from zero on the number line, and a distance can never be less than zero.

10. Error Analysis – Gus evaluated $-|-7|$ and wrote 7. Identify his mistake, then evaluate correctly. (Error Analysis)

11. Peak A is taller than peak B. Which expression is NOT equivalent to the difference in their heights? Circle one and explain. (Reason)

Ⓐ $|a - b|$ Ⓑ $|b - a|$ Ⓒ $|b| - |a|$ Ⓓ $|a| - |b|$

Answer: Ⓒ. When $a > b$, $|b| - |a|$ is negative, and a difference in heights (a distance) cannot be negative. Ⓐ and Ⓑ are equal; Ⓓ works only because both heights are positive.

12. A thermometer reads 98.5°F when the true temperature is 98.6°F. Write and evaluate an absolute value expression for the accuracy. (Apply)